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	AXI V810 Y-Axis Limit Sensor Adjustment Procedure	Effective Date	20 Oct 2016

This procedure is applicable to below models:

V810 STD	☒	S1	☒	S2
V810 XXL	☒	S1	☒	S2
V810 S2EX	☒			
V810 Mini	☐			

REVISION HISTORY

Rev	Effective Date	Description	Doc Originator
00	15 Jan 2013	First Release	ONG KEAN SHENG
01	04 Jan 2015	Second Release	MUHAMMED SHAMEEM
02	20 Oct 2016	Template Changes & Content Update	ONG KEAN SHENG

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1. Close the V810 software.
2. Login to the Atom PC and enter the computer name as shown below in Figure 1 (S1 Series) and Figure 2 (S2 Series)

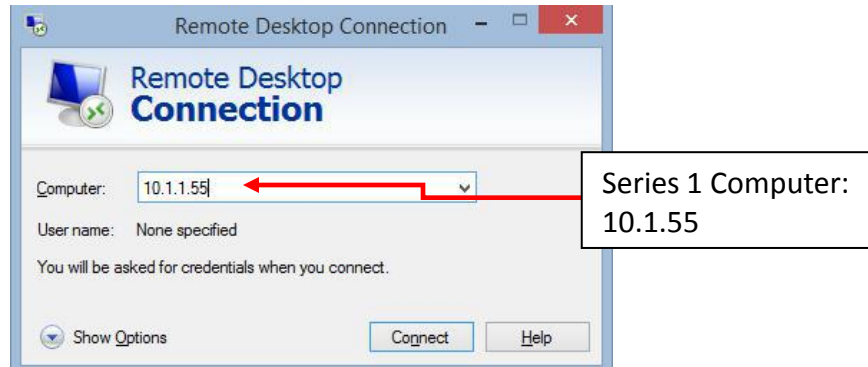


Figure 1 Series 1

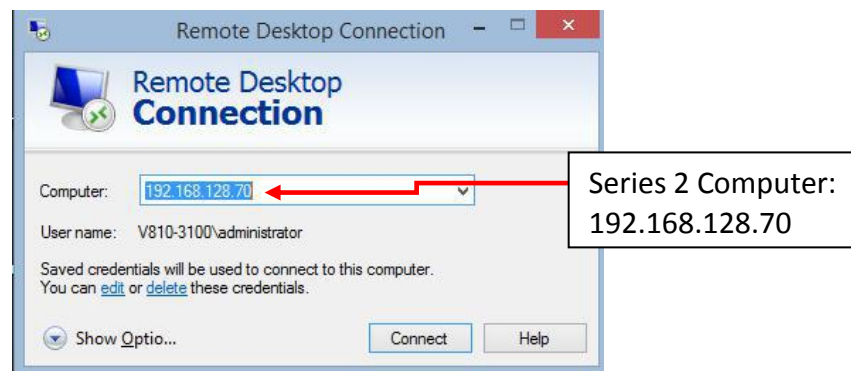


Figure 2 Series 2

3. Enter the password as "Please!" as shown in Figure 3.

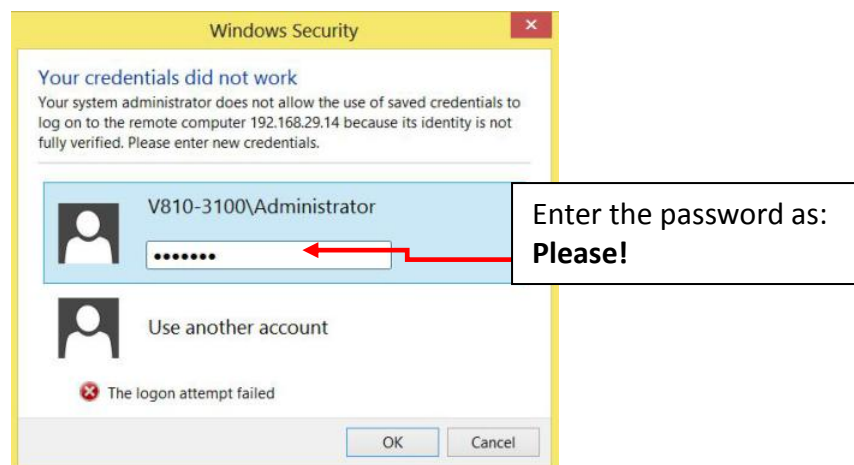


Figure 3 Password for Atom PC

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4. Go to the following directory: C:\Program Files\AXI System\v810\REV\bin and open the MC_XMP_NT.EXE as shown in Figure 4.

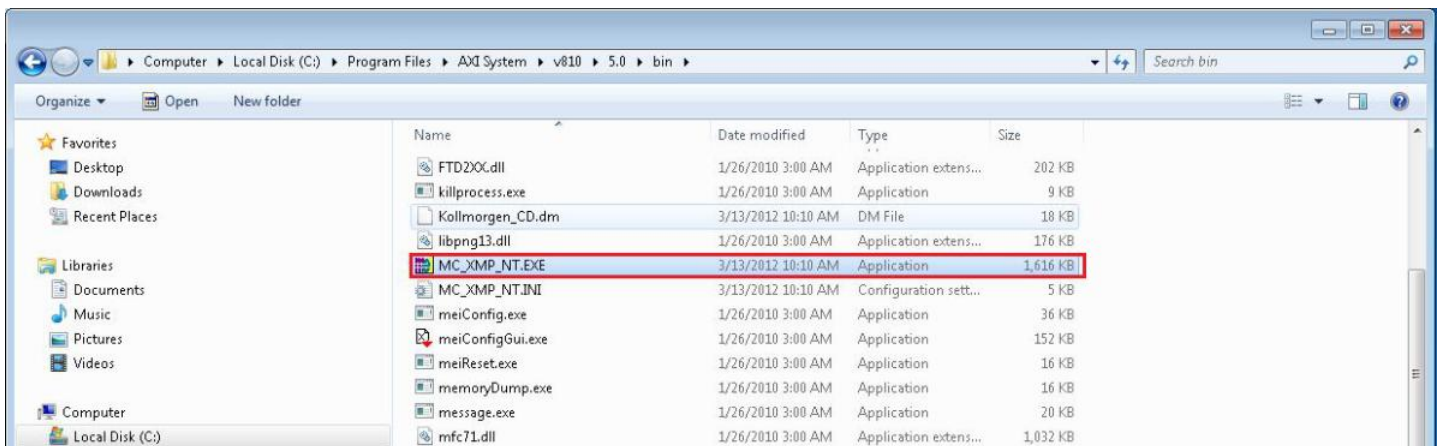


Figure 4 MC_XMP_NT.EXE

5. Ensure this 3 tab is open in the Motion Console tab as shown in Figure 5.

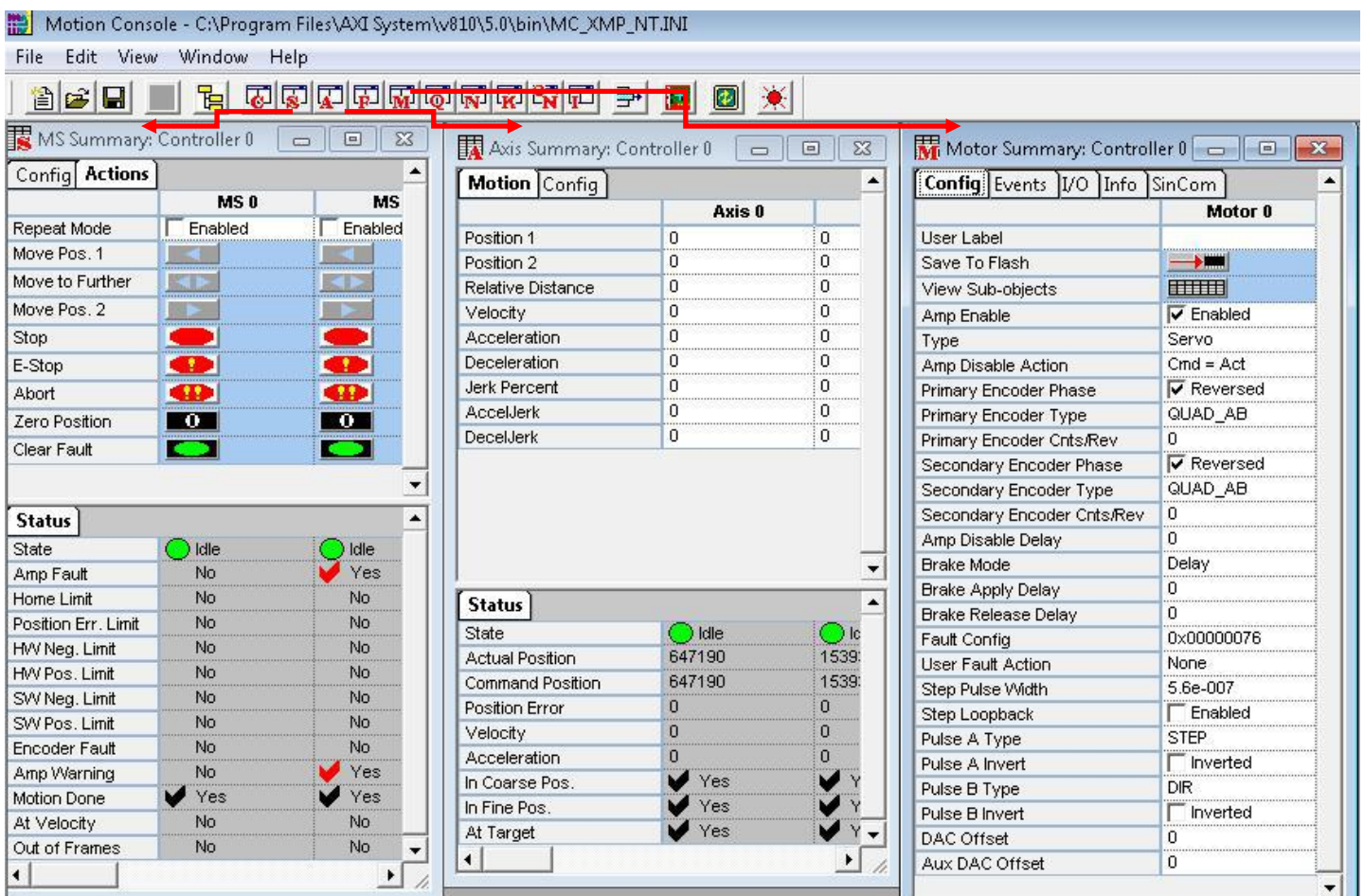


Figure 5 Motion Console tab

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6. Close the X-ray source and open the front access door.
7. Push the stage towards the minimum hard stop (front access door) as shown in below Figure 6.

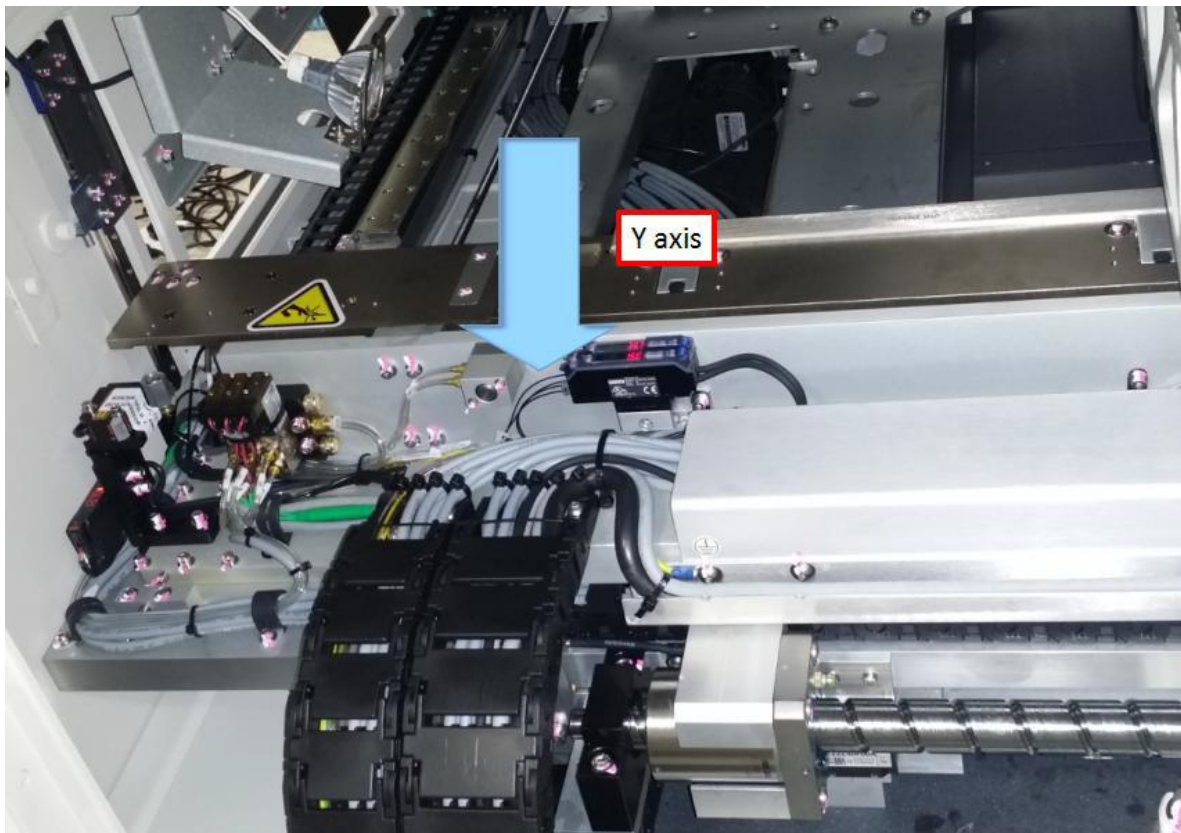


Figure 6 Y axis Minimum hard stop (V810 Std)

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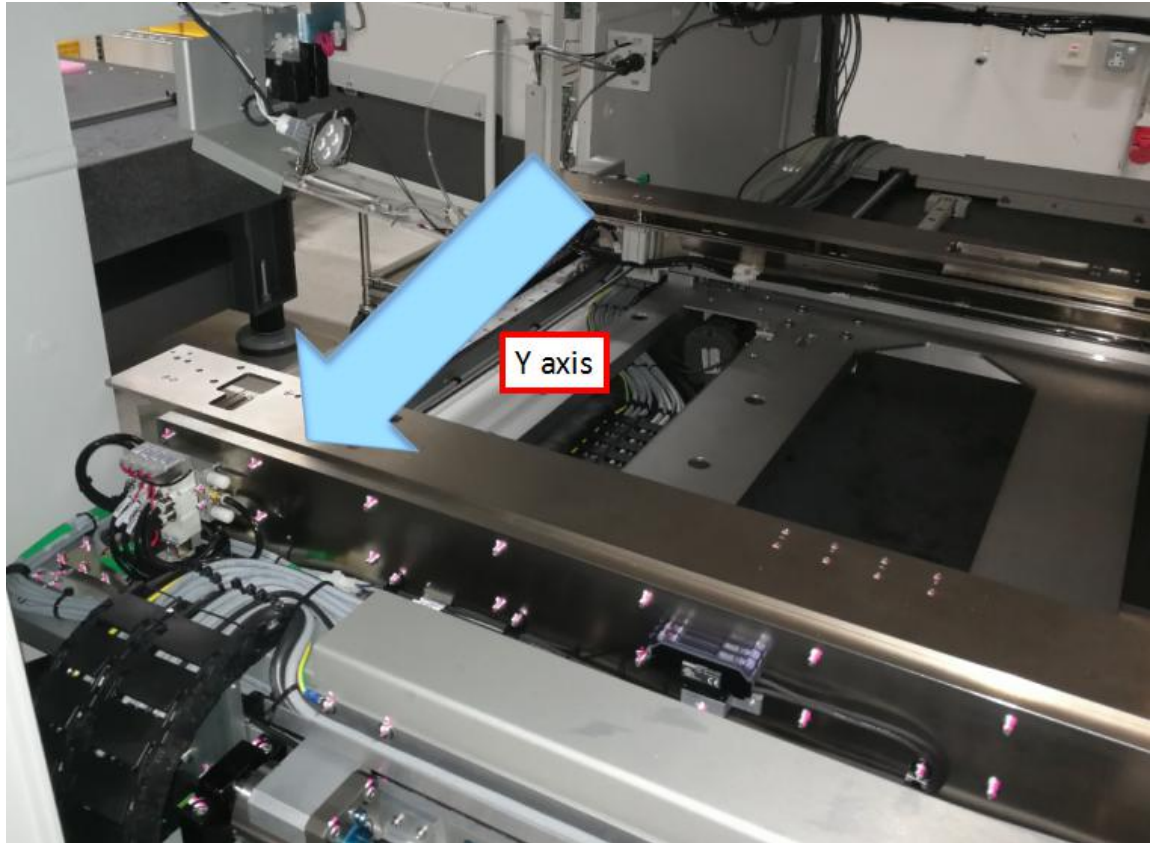


Figure 7 Y axis Minimum hard stop (V810 XXL)

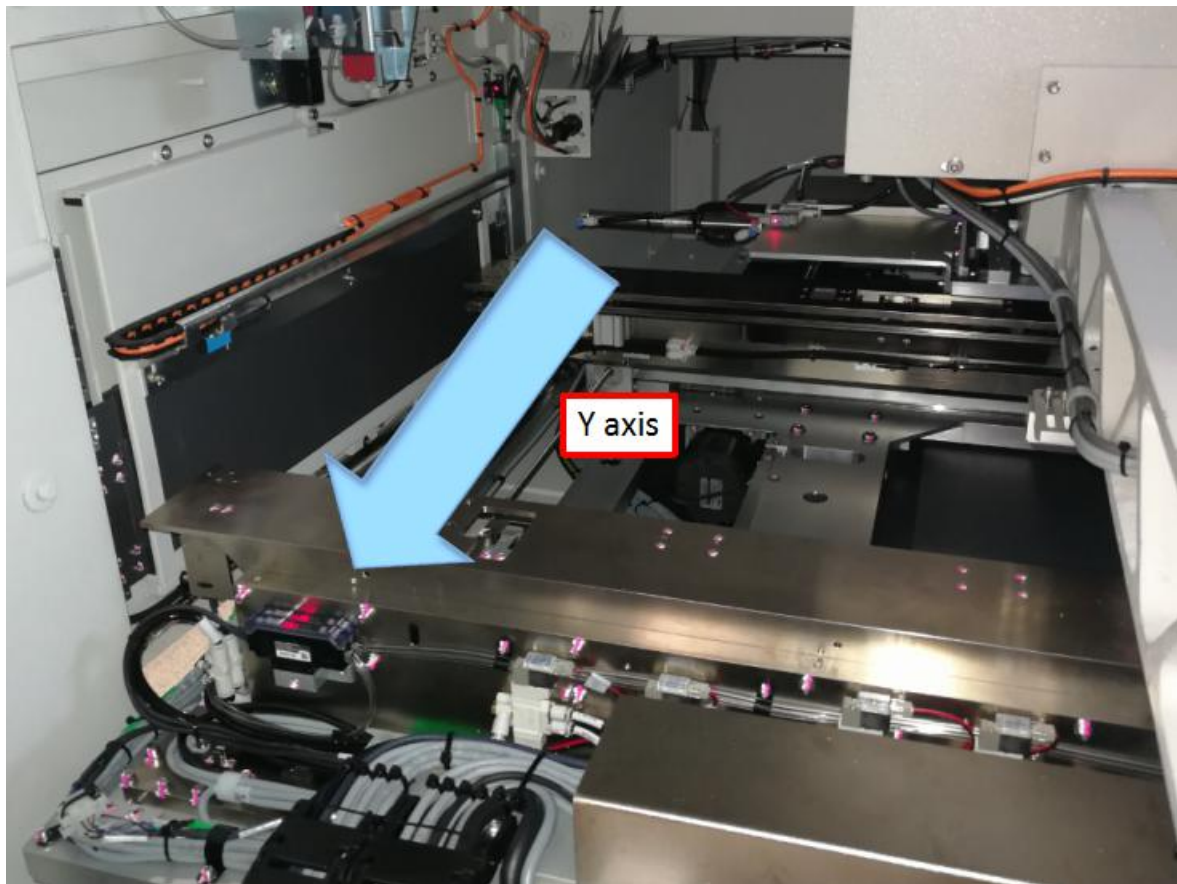


Figure 8 Y axis Minimum hard stop (V810 S2EX)

8. Reset all MS0, MS1 and MS2 at the MS Summary (Motion Supervisor Summary) as shown in Figure 7. Once reset for all the axis, the actual position for MS0, MS1 and MS2 will become zero as shown in Figure 8.

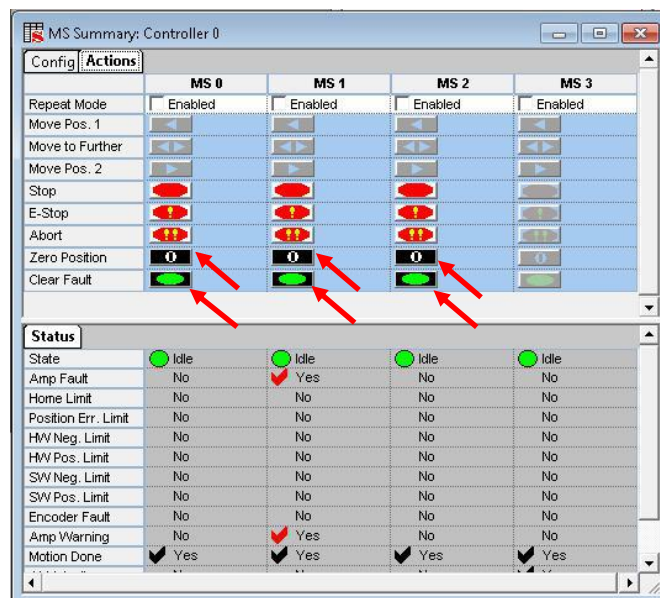


Figure 9 MS Summary (Motion Supervisor Summary)

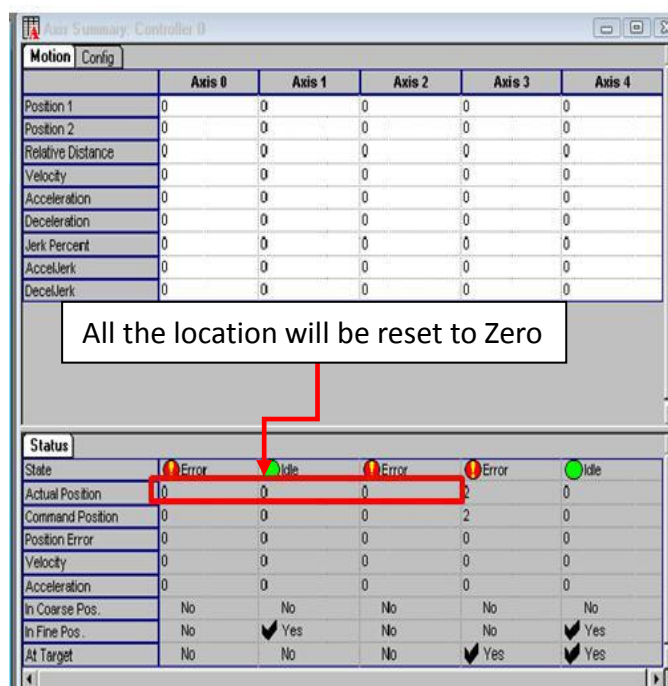


Figure 10 Axis Summary (Actual position)

9. Gently rotate the coupling for the Y-Axis stage as shown in the Figure 9 until the Actual Position and Command Position in Figure 10 become 5990.

Remarks: Acceptable range for the Actual Position and Command Position will be from 5970~5990.

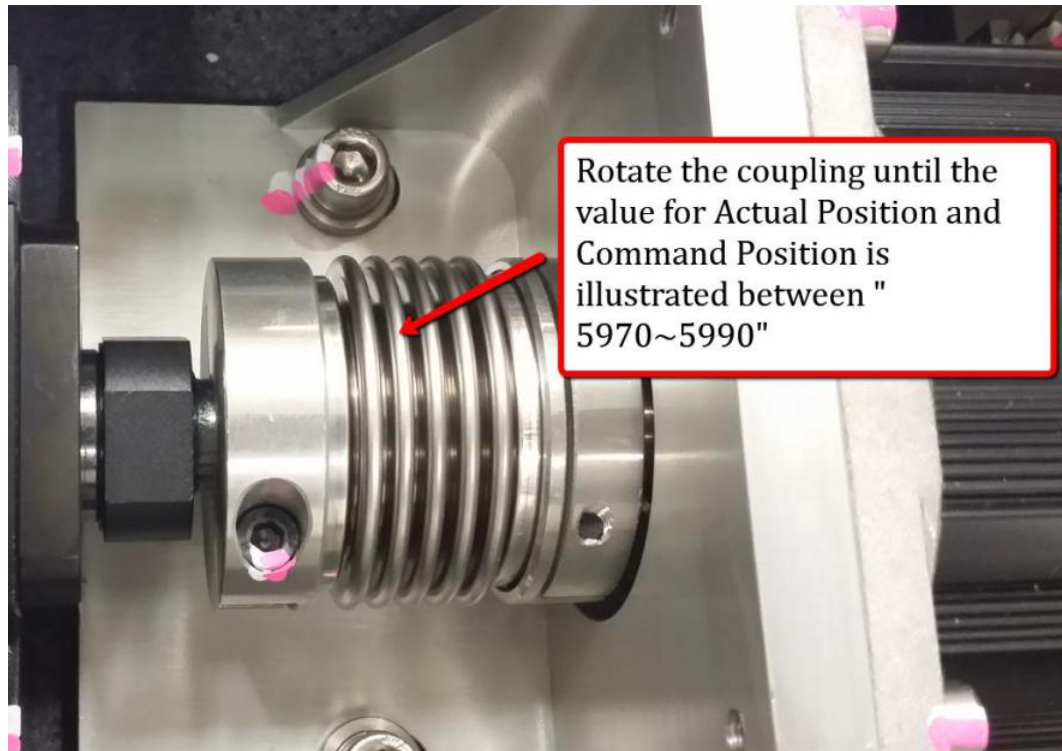
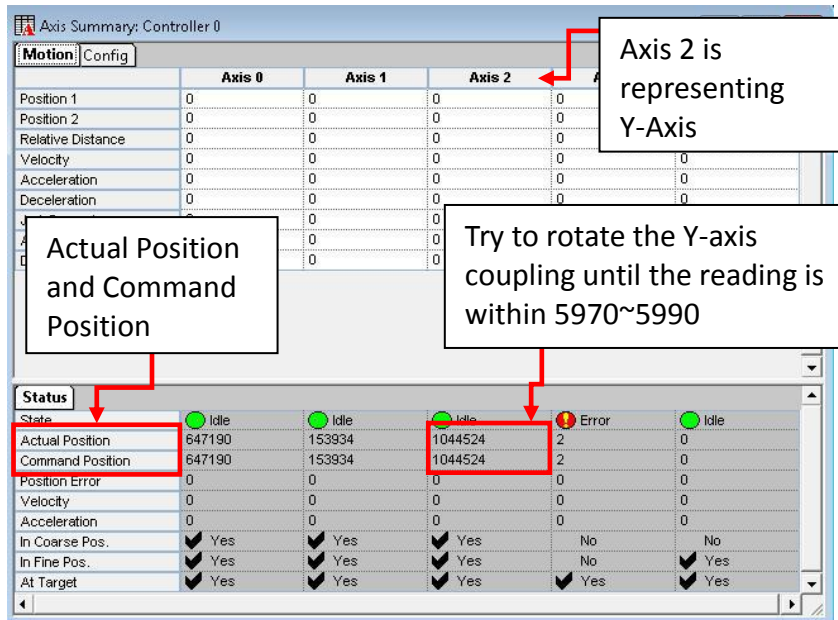


Figure 11 Y-axis coupling



	Axis 0	Axis 1	Axis 2
Position 1	0	0	0
Position 2	0	0	0
Relative Distance	0	0	0
Velocity	0	0	0
Acceleration	0	0	0
Deceleration	0	0	0
Actual Position	647190	153934	1044524
Command Position	647190	153934	1044524
Position Error	0	0	0
Velocity	0	0	0
Acceleration	0	0	0
In Coarse Pos.	✓ Yes	✓ Yes	✓ Yes
In Fine Pos.	✓ Yes	✓ Yes	✓ Yes
At Target	✓ Yes	✓ Yes	✓ Yes

Figure 12 Actual Position and Command Position

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10. Loosen the M3 screw of the limit switch as shown in Figure 11.

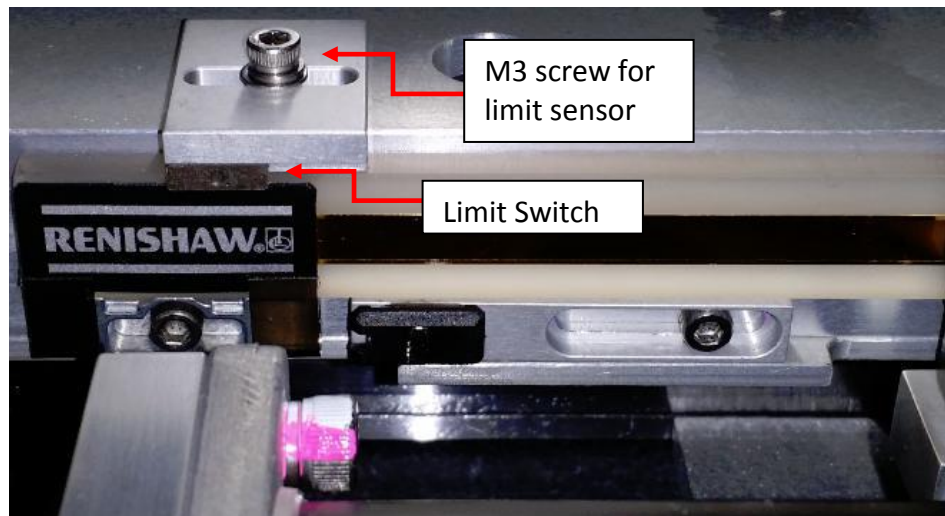
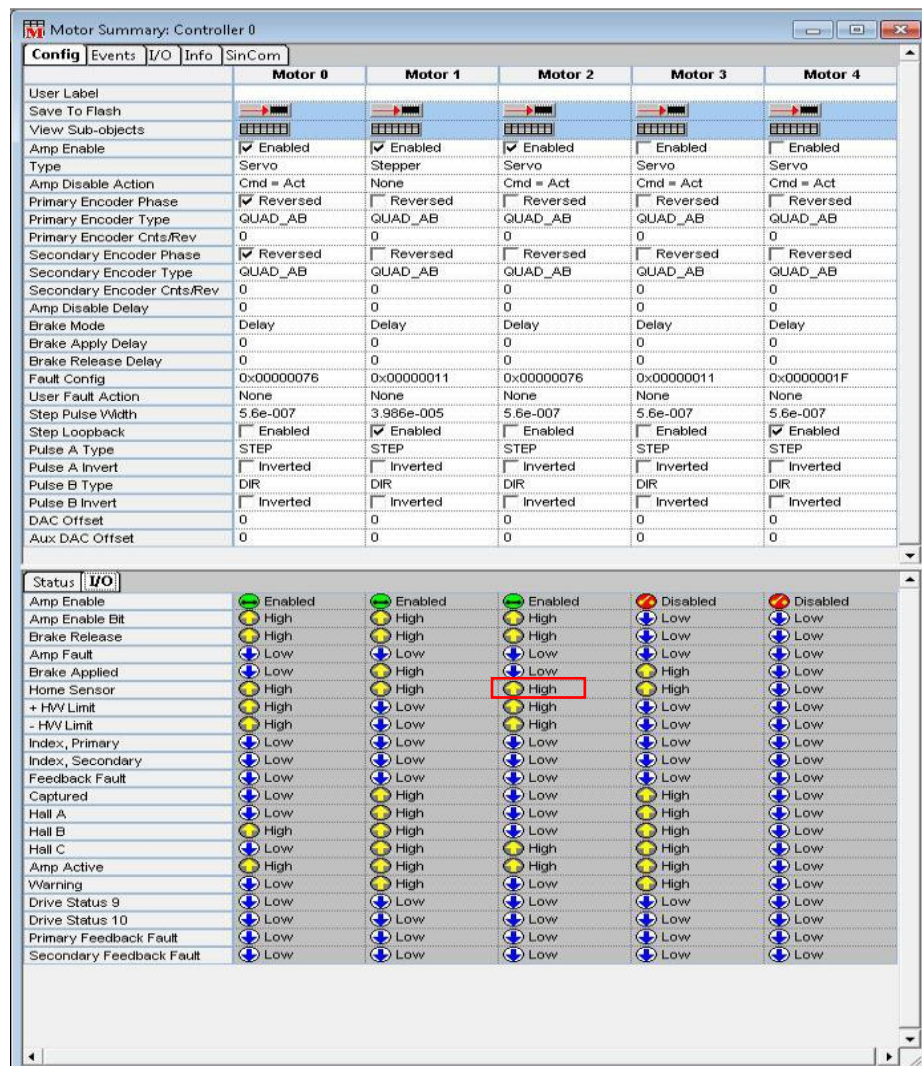


Figure 13 Limit Sensor for Y-Axis

11. Adjust the limit switch shown in Figure 11 until the Home Sensor status become “High” at the Motor Summary as shown in Figure 12 and Figure 13.

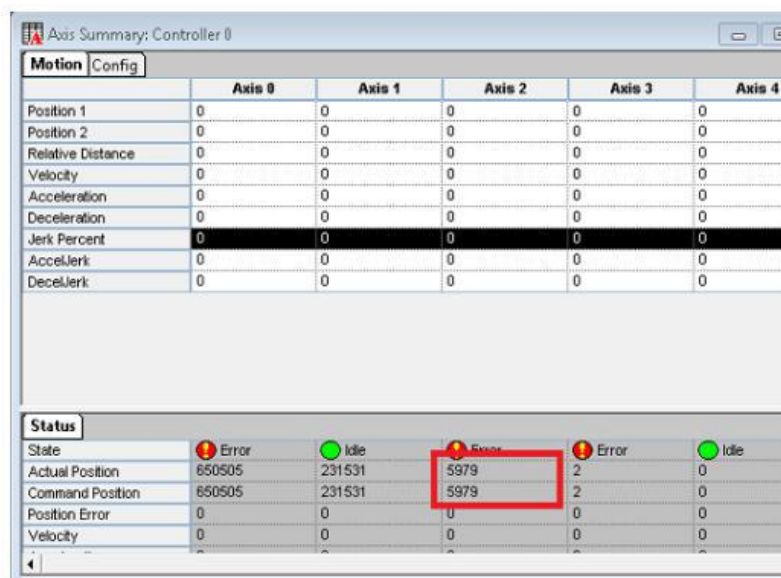


The screenshot shows the 'Motor Summary' window for Controller 0. It has tabs for Config, Events, I/O, Info, and SinCom. The 'Config' tab is active, showing settings for five motors. The 'Status' tab is also visible, showing the current status of various sensors and signals. The 'Home Sensor' status for Motor 2 is highlighted with a red box, showing a 'High' status.

	Motor 0	Motor 1	Motor 2	Motor 3	Motor 4
User Label					
Save To Flash					
View Sub-objects					
Amp Enable	☒ Enabled	☒ Enabled	☒ Enabled	☒ Enabled	☒ Enabled
Type	Servo	Stepper	Servo	Servo	Servo
Amp Disable Action	Cmd = Act	None	Cmd = Act	Cmd = Act	Cmd = Act
Primary Encoder Phase	☒ Reversed	☐ Reversed	☐ Reversed	☐ Reversed	☐ Reversed
Primary Encoder Type	QUAD_AB	QUAD_AB	QUAD_AB	QUAD_AB	QUAD_AB
Primary Encoder Cnts/Rev	0	0	0	0	0
Secondary Encoder Phase	☒ Reversed	☐ Reversed	☐ Reversed	☐ Reversed	☐ Reversed
Secondary Encoder Type	QUAD_AB	QUAD_AB	QUAD_AB	QUAD_AB	QUAD_AB
Secondary Encoder Cnts/Rev	0	0	0	0	0
Amp Disable Delay	0	0	0	0	0
Brake Mode	Delay	Delay	Delay	Delay	Delay
Brake Apply Delay	0	0	0	0	0
Brake Release Delay	0	0	0	0	0
Fault Config	0x00000076	0x00000011	0x00000076	0x00000011	0x0000001F
User Fault Action	None	None	None	None	None
Step Pulse Width	5.6e-007	3.986e-005	5.6e-007	5.6e-007	5.6e-007
Step Loopback	☐ Enabled	☒ Enabled	☐ Enabled	☐ Enabled	☒ Enabled
Pulse A Type	STEP	STEP	STEP	STEP	STEP
Pulse A Invert	☐ Inverted	☐ Inverted	☐ Inverted	☐ Inverted	☐ Inverted
Pulse B Type	DIR	DIR	DIR	DIR	DIR
Pulse B Invert	☐ Inverted	☐ Inverted	☐ Inverted	☐ Inverted	☐ Inverted
DAC Offset	0	0	0	0	0
Aux DAC Offset	0	0	0	0	0

	Motor 0	Motor 1	Motor 2	Motor 3	Motor 4
Amp Enable	Enabled	Enabled	Enabled	Disabled	Disabled
Amp Enable Bit	High	High	High	Low	Low
Brake Release	High	High	High	Low	Low
Amp Fault	Low	Low	Low	Low	Low
Brake Applied	Low	High	Low	High	Low
Home Sensor	High	Low	High	Low	Low
+ H/V Limit	High	Low	High	Low	Low
- H/V Limit	High	Low	High	Low	Low
Index, Primary	Low	Low	Low	Low	Low
Index, Secondary	Low	Low	Low	Low	Low
Feedback Fault	Low	Low	Low	Low	Low
Captured	Low	High	Low	High	Low
Hall A	Low	High	Low	High	Low
Hall B	High	High	Low	High	Low
Hall C	Low	High	High	High	Low
Amp Active	High	High	High	High	Low
Warning	Low	High	Low	High	Low
Drive Status 9	Low	Low	Low	Low	Low
Drive Status 10	Low	Low	Low	Low	Low
Primary Feedback Fault	Low	Low	Low	Low	Low
Secondary Feedback Fault	Low	Low	Low	Low	Low

Figure 14 Motor Summary (Home Sensor)



The screenshot shows the 'Axis Summary' window for Controller 0. It has tabs for Motion and Config. The 'Motion' tab is active, showing the current position and velocity of five axes. The 'Status' tab is also visible, showing the current status of various signals. The 'Home Sensor' status for Axis 2 is highlighted with a red box, showing a 'High' status.

	Axis 0	Axis 1	Axis 2	Axis 3	Axis 4
Position 1	0	0	0	0	0
Position 2	0	0	0	0	0
Relative Distance	0	0	0	0	0
Velocity	0	0	0	0	0
Acceleration	0	0	0	0	0
Deceleration	0	0	0	0	0
Jerk Percent	0	0	0	0	0
AccelJerk	0	0	0	0	0
DecelJerk	0	0	0	0	0

	Axis 0	Axis 1	Axis 2	Axis 3	Axis 4
State	Error	Idle	Error	Error	Idle
Actual Position	650505	231531	5979	2	0
Command Position	650505	231531	5979	2	0
Position Error	0	0	0	0	0
Velocity	0	0	0	0	0

Figure 15 Axis Summary (Home Sensor)

12. Once the limit switch condition is acquired, tighten back the M3 screw for the limit sensor screw.
13. Pull the Y-Axis stage to the minimum hard stop (front access door).
14. Reset again the Axis 2 become zero as shown in Figure 14 and ensure the actual position for the Axis 0, Axis 1 and Axis 2 become zero as shown in Figure 15.

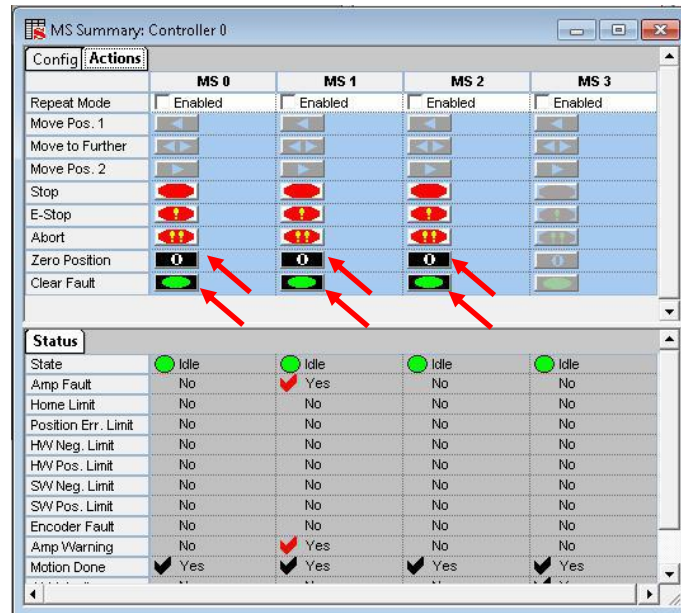


Figure 16 MS Summary

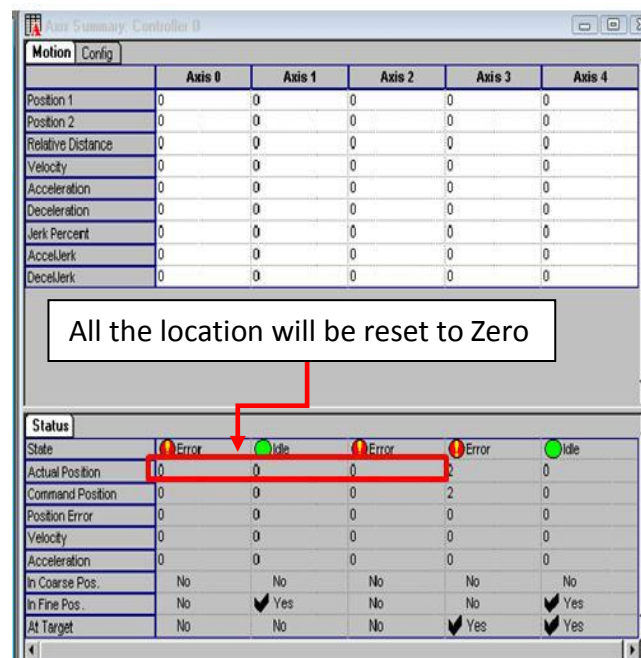


Figure 17 Axis Summary (Actual Position)

15. Slowly rotate the Y-Axis coupling until status become “Low” at indicated at the “Home Sensor” as shown in Figure 16.

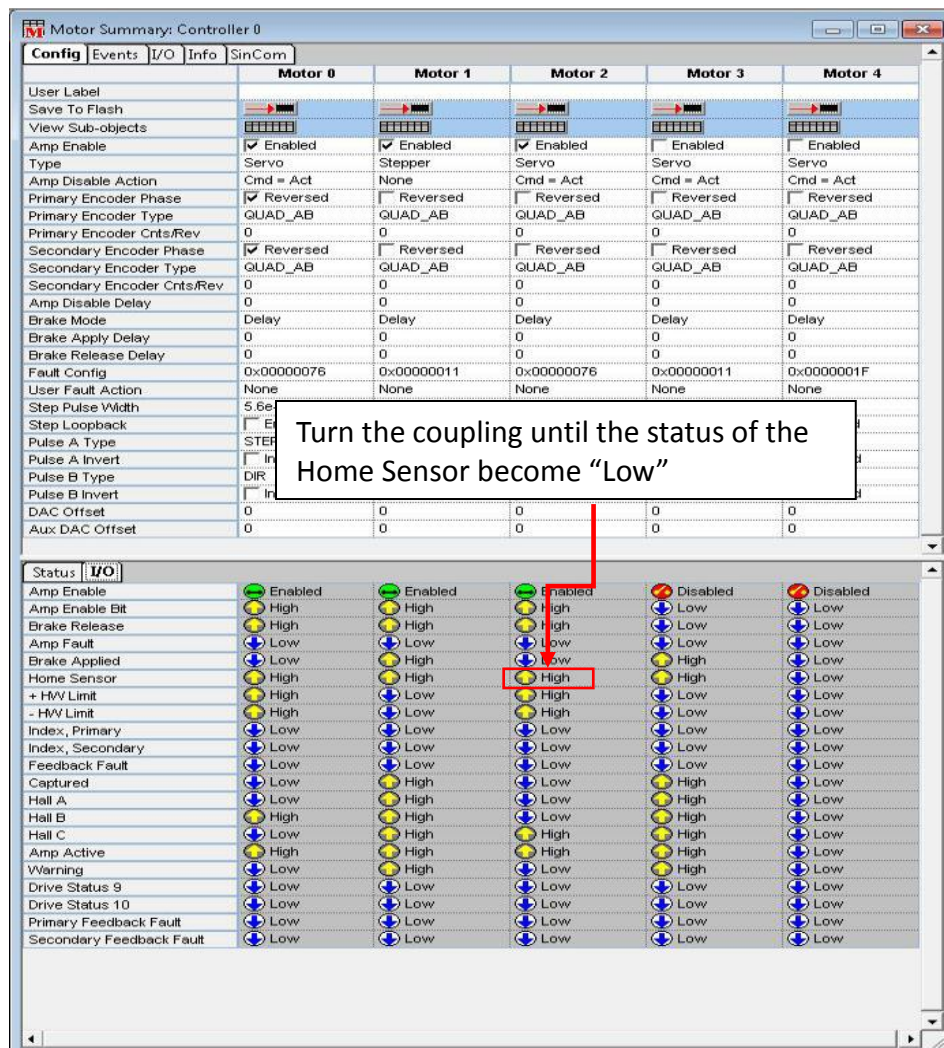


Figure 18 Motor Summary

16. Once the status becomes “Low”, Y-Axis stage should not move anymore.

17. Refer to the actual (From Axis Summary) to verify the limit sensor position. May refer to below table for counter reference.

Y-Axis Actual position (Axis 2 at Axis summary)	Expected result at Motor 2 (Motor summary)
5000~5200	Home Sensor Low
5700~5990	Home Sensor High

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18. Transition signal from low to high will happen at the range of 5000-6000 counts. If the result is not able to achieved as above table (Low at 5000-5200 and High at 5700-5990), please do repeat from Step 5-17.
19. Exit the Motion Console software.
20. Push the XY-stage to the middle of the machine and close the front access door.
21. Open the V810 software. Go to the Services->Subsystem Status & Control-> Digital IO tab and Click Initialize.
22. Go to the Panel Positioner tab and click Initialize. The limit sensor status for the Y-axis should be clear. Disable the Motion for the X-axis and Y-axis as shown in Figure 17. Close the X-ray source and open the front door.

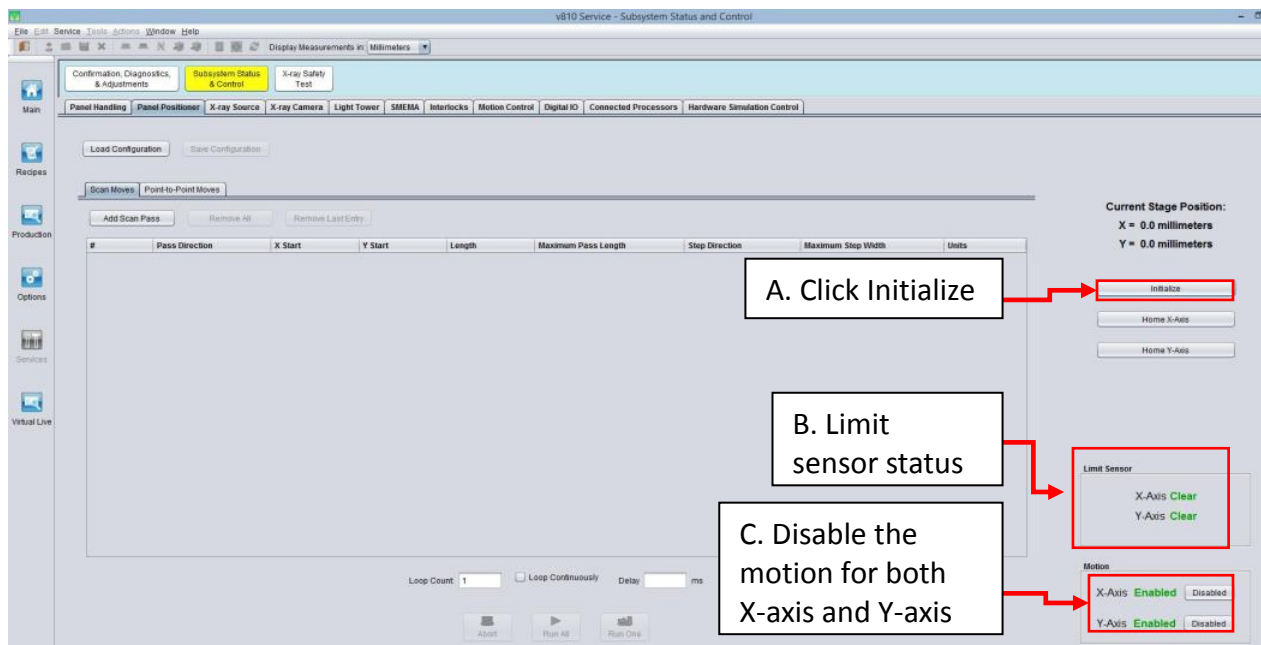
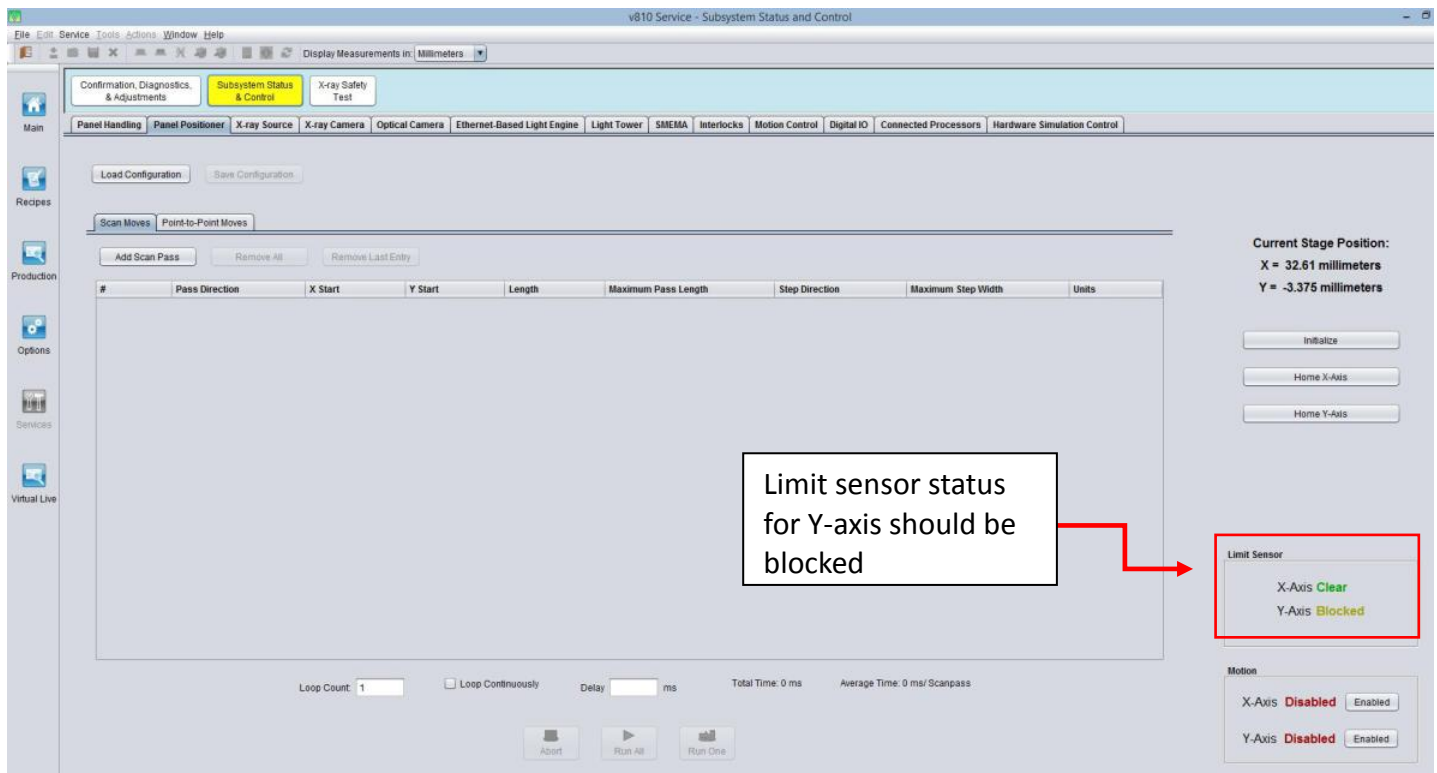


Figure 19 Panel Positioner

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23. Push the stage to the minimum hard stop (left outer barrier side) and the status of the X-axis limit sensor should be blocked as shown in Figure 18.



24. If verification not be able to achieved, repeat Step 5-23.